

Auteur: Siebe Mees
 Studierichting: Informatica
 Oefeningenreeks 2B nr. 6.7

$$\begin{aligned}
 & 4 \operatorname{Bgtan} \frac{1}{5} - \operatorname{Bgtan} \frac{1}{239} = \frac{\pi}{4} \\
 & = 2 \operatorname{Bgtan} \frac{\frac{1}{5} + \frac{1}{5}}{1 - \frac{1}{5} \cdot \frac{1}{5}} - \operatorname{Bgtan} \frac{1}{239} \\
 & = 2 \operatorname{Bgtan} \frac{10}{24} - \operatorname{Bgtan} \frac{1}{239} \\
 & = \operatorname{Bgtan} \frac{\frac{10}{24} + \frac{10}{24}}{1 - \frac{10}{24} \cdot \frac{10}{24}} - \operatorname{Bgtan} \frac{1}{239} \\
 & = \operatorname{Bgtan} \frac{120}{119} - \operatorname{Bgtan} \frac{1}{239} \\
 & = \operatorname{Bgtan} \frac{\frac{120}{119} - \frac{1}{239}}{1 + \frac{120}{119} \cdot \frac{1}{239}} \\
 & = \operatorname{Bgtan} \frac{\frac{28441}{28441} - \frac{28441}{120}}{\frac{28441}{28441} + \frac{28441}{28441}} \\
 & = \operatorname{Bgtan} 1
 \end{aligned}$$

References